



DEPARTMENT OF CIVIL ENGINEERING

Module: 2

Introduction to sustainable development goals:

1. **NO POVERTY:** End poverty in all its forms everywhere.
2. **ZERO HUNGER:** End hunger, achieve food security and improved nutrition and promote sustainable agriculture
3. **GOOD HEALTH & WELL-BEING:** Ensure healthy lives and promote well-being for all at all ages
4. **QUALITY EDUCATION:** Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
5. **GENDER EQUALITY:** Achieve gender equality and empower all women and girls
6. **CLEAN WATER AND SANITATION:** Ensure availability and sustainable management of water and sanitation for all
7. **AFFORDABLE & CLEAN ENERGY:** Ensure access to affordable, reliable, sustainable and modern energy for all
8. **DECENT WORK AND ECONOMIC GROWTH:** Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all
9. **INDUSTRY, INNOVATION AND INFRASTRUCTURE:** Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation
10. **REDUCED INEQUALITIES:** Reduce inequality within and among countries
11. **SUSTAINABLE CITIES AND COMMUNITIES:** Make cities and human settlements inclusive, safe, resilient and sustainable
12. **SUSTAINABLE CONSUMPTION AND PRODUCTION:** Ensure sustainable consumption and production patterns
13. **CLIMATE ACTION:** Take urgent action to combat climate change and its impacts
14. **LIFE BELOW WATER:** Conserve and sustainably use the oceans, seas and marine resources for sustainable development
15. **LIFE ON LAND:** Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss
16. **PEACE, JUSTICE AND STRONG INSTITUTIONS:** Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels
17. **PARTNERSHIPS FOR THE GOALS:** Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development.



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Smart city concept

A **smart city** is a technologically modern urban area that uses different types of electronic methods and sensors to collect specific data. Information gained from that data is used to manage assets, resources and services efficiently; in return, that data is used to improve operations across the city.

This includes data collected from citizens, devices, buildings and assets that is processed and analyzed to monitor and manage traffic and transportation systems, power plants, utilities, water supply networks, waste, Criminal investigations, information systems, schools, libraries, hospitals, and other community services.

Smart cities are defined as smart both in the ways in which their governments harness technology as well as in how they monitor, analyze, plan, and govern the city.

The Government of India, amidst a lot of fanfare, launched the **Smart Cities Mission in 2015**. The objective of this mission is to develop inclusive and sustainable citizen friendly cities enhancing the quality of life of citizens. The focus of the smart city in India is on investing in the development and infrastructure of Tier 1 and 2 cities bringing them at the same level as global smart cities.

Smart City Development Solutions

The solutions as part of Smart Cities Mission are aimed at achieving comprehensive development. Some of these solutions are-

- Availability of easy to access transportation options
- Promotion of mixed land use
- Development of open, recreational spaces
- Expanding excellent housing opportunities
- Inclusive and sustainable development
- Seamless Integration of compact areas with the neighbouring areas.

The core elements of Smart City Infrastructure include-

- ✓ Assured Supply of Electricity
- ✓ Adequate Supply of Water
- ✓ Solid Waste Management
- ✓ Sanitation



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- ✓ Efficient Urban Mobility
- ✓ Affordable Housing
- ✓ Public Transport
- ✓ Digitalisation
- ✓ IT Connectivity
- ✓ Sustainable Environment
- ✓ Good Governance
- ✓ Health and Education
- ✓ Security of the Citizens

The aim is to create a self-sufficient city that takes care of its every necessity and citizen needs effectively and efficiently. It results in the citizens becoming better contributors for a successful society. It also results in contributing towards an effective growth and development of the city.

Clean city concept

The term 'green city' refers to a city of greenery – that of trees, plants, etc. while the term 'clean city' refers to a city that is free from pollution and environment degradation. Both these terms are relative in nature. A city that has many trees than buildings all over will be free from environmental problems as trees help to purify the air and maintain the cycle of nature. When a city is kept free from pollution and virtuous it attracts the attention of tourists and visitors and earns a prominent place in the lives of its citizens.

It's our own interest to grow more trees for maintaining the balance in nature.

The CLEAN-India Programme:-

India has a population of over one billion, of which almost 300 million people are living in 600 towns and cities. It is unfortunate that, as a result of stressed environmental conditions, most of these towns and cities are unable to keep the pace for development. Water pollution, Depletion of Groundwater, inadequate sanitation, open dumping of waste, and loss of forest cover are the problems. These impact on the health of the people and also does an economic



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impact on the country. Similarly, water diseases like diarrhea, jaundice, and cholera are increasing daily on the basis of pollution done by us and are affecting both human health and economic productivity.

CLEAN-India Mission:-

The CLEAN-India program aims to deploy responsibility to the people for the assessment of environmental consequences and improvement in all major cities and states in India through schools and NGOs linked with governments and other private organizations.

Things are not under control when there is the absence of proper garbage and sewage treatment plants. In this situation, the cities in India will soon become unfriendly.

The NGOs are made up for helping spreading awareness about the clean environment they also monitor things which are carried on or things which are happening which affects the environment.

They, stop unnecessary things from happening like Cutting down of trees for silly things or for the projects (i.e Metro Projects) which are done for the development of the public sector, resulting in cutting down of trees.

Things like this happen all our life, but how to react to it is the main thing for helping protect the Mother Nature.

If we take nature for granted and not take necessary steps then, our planet will perish and so will we.

Safe city concept

The safe city concept has been developed precisely to help government stakeholders, city mayors, and police departments mitigate these challenges. Leveraging the internet of things (IoT) and the connectivity now found in core security and safety technologies, safe city solutions are providing a range of systems including predictive analytics and big data, real-time response procedures and emergency response systems. These solutions are enabling governments and police departments to better protect their citizens from everything from terrorist attacks to natural



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disasters. They are also supporting wider city organizations such as public health, fire and rescue, border control and social services to better serve their citizens.

The safe city concept is based on a consolidated ICT platform which combines public-safety information of different types and from different sources, obtained through sensors and multi-agency collaboration.

Safe cities are an essential pillar supporting the development of smart cities throughout the world. They provide the security and safety required to protect citizens from crime and terrorism as well as mitigate, as much as possible, the impact of natural disasters and other threats. A successful safe city solution should support a city's security agencies, fire and rescue departments, public health and social service departments before, during and after an event, and integrate the disparate technologies and government departments responsible for citizen safety. The safe city concept is based on a consolidated ICT platform which combines public-safety information of different types and from different sources, obtained through sensors and multi-agency collaboration.

Four major requisites

- ***Prevention: It provides reliable and comprehensive security measures to predict threats and hazardous situations.***

City authorities must be able to take measures to prevent threats from occurring in the first place. Simulation and forecasting technology, based on big data mining, can help the relevant authorities to predict public threats and support police and military assets to prevent the event before it begins.

- ***Detection: It aids to public-safety organizations in collecting, sharing and analyzing data more effectively to provide early warnings and raise situational awareness.***

Sensor systems in the city will proactively gather information. These sensors may include video surveillance cameras, CBRNE (Chemical, biological, radiological and nuclear) sensors, gunshot-detection sensors, and weather sensors. The type of sensor used is determined by the scope of the safe city project and how much inter-operability is required. Information can be used to provide early detection and alerts when events occur. Increasingly, sensors include 'listening' to social media for relevant posting by netizens.



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- **Response:** *It enables the key organizations in the city to react to security threats in real time.*

City authorities must be able to prevent an adverse event from escalating. Safe city projects enable an effective response by using a consolidated ICT platform to provide a common operational picture to all relevant agencies, including law enforcement, public health, fire and rescue services; and to allow the critical communication systems of these agencies to inter-operate. This helps to raise the situational awareness for each of the responding agencies, and their command, control and coordination.

- **Recovery:** *It provides post-event examination and analysis, identify victims and provide assistance in rescue actions.*

After an event, city authorities must be able to quickly examine and analyze all data received from the relevant sensor systems. This supports the subsequent search for suspects as well as the process of building a case based on incriminating evidence. Information can also be used to facilitate rescue actions, and the identification and assistance processes for both victims and survivors.

Urban air pollution management:

“Air Pollution is the release of pollutants such as gases, particles, biological molecules, etc. into the air that is harmful to human health and the environment.”

[Urbanization](#) and rapid industrialization have benefited mankind and made the life of humans easier and comfortable. However, both urbanization and industrialization also pose harm to mankind, the top of which is air pollution. Urban air pollution refers to the air pollution in and around cities. Denser populations experience more urban air pollution.

Air pollution affects human health as well as the climate of an area. According to [World Health Organization \(WHO\)](#) 4.2 million deaths every year occur as a result of exposure to ambient (outdoor) air pollution. Although there are some



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natural sources of urban air pollution, most of the sources are anthropogenic and largely depend on the activities of people.

Natural sources of urban air pollution include volcanic eruptions, thunders, dust from the earth's surface and naturally occurring particulate matter. Nevertheless, major factors that contribute to urban air pollution are anthropogenic activities, including transportation, domestic use of fossil fuels, industrialization, power generation, combustion and agriculture and beauty products.

Urban Air Pollution Sources

Transportation

The use of private vehicles (particularly older, diesel models) is the major source of urban air pollution. The US Environmental Protection Agency estimates that about 75% of VOC emissions (by weight) come from transportation. About one-quarter of particulate matter in the air is due to vehicles.

Domestic use of fossil fuels

Half of the world population still relies on solid fuels for cooking and heating. These fuels, including wood, charcoal and coal, are burned in inefficient stoves that release large quantities of health-damaging particulate matter and climate warming pollutants into the nearby environment. In addition, it is estimated that 1.2 billion people light their home with kerosene lamps additionally contributing to air pollution and increasing risk of respiratory and cardiovascular diseases.

Industrialization

Industrialization is also a major contributor to urban air pollution as the area having industries particularly show poor air quality. Factories release many toxic gases due to the burning of fossil fuels and the use of chemicals. These gases react



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with each other and with other atmospheric constituents. It is estimated that around 80 different toxins can be found in the air emitted by factories, from asbestos and dioxin to lead and chromium.

Power generation

With increased population, there is an increased energy demand. To fulfill that demand, fossil fuels are tremendously being used to generate energy as they are cheap and readily. The coal powered power plants are a major source of urban air pollution.

Combustion and agriculture

Combustion of material is an activity that releases toxic gases in the atmosphere and contributes to urban air pollution. Combustion releases CO_2 and incomplete combustion releases CO . Both of these gases lead to urban air pollution. Agriculture activities also release other gases in the atmosphere such as NO_2 and Methane (CH_4).

Beauty Products

A recent study suggests that the use of beauty products also contributes to urban air pollution. Most of the cosmetics and perfumes contain VOCs, which are released during their use and contribute to urban air pollution. With more population, the use of these beauty products is increasing and so the pollution.

Air Pollutants

The quality of urban air is indicated by the quantity of certain pollutants in the air, like Ozone (O_3), Particulate Matter (PM_{10} , $\text{PM}_{2.5}$), Sulfur Oxides (SO_x), Nitrogen Oxides (NO_x), Carbon monoxide (CO) and Volatile Organic Compounds (VOCs).



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Air Pollution Control

Following are the measures one should adopt, to control air pollution:

1. Using public transports

Using public transport is a sure short way of contributing to less air pollution as it provides with less gas and energy, even carpools contribute to it. In addition to less release of fuels and gas, using a public transport can also help in saving money. Lesser vehicles on roads will contribute to lesser emissions. Using public transport will help in:

- Reducing the overall emissions
- Number of vehicles reduced on the roads
- Lesser traffic congestion episodes
- It is pocket friendly alternative

2. Turn off the lights when not in use

The energy that the lights take also contribute to air pollution, thus less consumption of electricity can save energy. Use energy saving fluorescent lights to help the environment.

3. Recycle and Reuse

The concept of recycle and reuse is not just conserve resources and use them judiciously but also is helpful for air pollution as it helps in reducing pollution emissions. The recycled products also take less power to make other products.

4. No to plastic bags

The use of plastic products could be very harmful to the environment as they take a very long time to decompose, due to their material made up of oil. The use of paper bags instead is a better alternative as they decompose easily and are recyclable.



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5. Reduction of forest fires and smoking

The collecting of garbage and getting it on fire in dry seasons or dry leaves catching fires is a huge factor for causing air pollution, moreover smoking also causes air pollution and causes the air quality to worsen along with obviously damaging one's health.

6. Use of fans instead of Air Conditioner

The usage of AC's takes a lot of energy and emits a lot of heat which is bad for the environment. AC's also take a lot of power and energy to work as compared to fans.

7. Use filters for chimneys

The gas that is emitted from fireplaces in homes and factories are extremely dangerous for air pollution and harms the air quality severely. The use of filters should be used at least if the consumption couldn't be lessened, this will help to reduce the effect of harmful gases absorbing in the air.

8. Avoid usage of crackers

The use of crackers during festivals and weddings is sadly one of the biggest contributors to air pollution, leading to a layer of smog which is extremely harmful for health. So, practice of no crackers should be implemented.

9. Avoid using of products with chemicals

Products that use the chemicals in their usage or smell strongly, like paints or perfumes should be used less or outside the house. There can also be an alternative to use products with low chemical content and organic properties.

10. Implement Afforestation

Last but not the least, plant and grow as many trees as possible. The practice of planting trees provides a lot of benefits to the environment and helps with the release of oxygen



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Urban flood control:

Urban flooding describes the flooding of land or property within a constructed environment, especially in locations that are more densely inhabited (like cities), as a result of excessive rainfall that exceeds the capacity of drainage systems.

Measures to prevent Urban Flooding

1. Need for Climate Resilient Infrastructure

- Given the unprecedented rate of climate change-related severe disaster in recent times, urban areas must have a climate-resilient infrastructure.
- According to UN projections, by 2050 more than 68% of the world's population could be concentrated in urban areas.
- Additionally, daily life in urban areas is highly dependent on certain critical services and products provided by critical infrastructures (CIs).
- Therefore, it is especially relevant to understand how Climate Change affects urban CIs in order to develop mechanisms to improve their capacity to handle crises derived from CC.
- In this context, resilience-based strategies provide a holistic approach, considering both predictable and unpredictable threats.
- Innovative approaches like Sponge Cities- wetland restoration, flushing systems using collected rooftop water, bioswales, and public spaces as flexible water retention facilities can be applied to Indian urban areas.
 - Other such methods include permeable material for roads and pavement, green roofs and harvesting systems in buildings.



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2. Early Warning Systems and Communication

- Dissemination of flood warnings must be carried out, using a wide range of latest technologies.
- This would help in giving real-time data where traditional systems fail.
- Tools such as predictive precipitation modelling can help do that and are also able to link it with the adaptive capacity of urban land use.

3. Design and Management of Urban Drainage System

- Watershed management and emergency drainage plan should be clearly enunciated in policy and law.
- Proper management of the drainage system is necessary to ensure that the water does not get stored in one place.
- Solid waste increases hydraulic roughness, causes blockage and generally reduces flow capacity.
- These drains need to be cleaned on a regular basis to permit the free flow of water.
- Vulnerability analyses and risk assessments should form part and parcel of city master plans.

4. Rainwater Harvesting

- Due to urbanization, groundwater recharge has decreased and the peak runoff from rainfall and consequent flooding has increased.
- It will serve the twin purposes of lowering the peak runoff and raising the groundwater table.
- Many municipal corporations in India have already made rainwater harvesting compulsory.

5. Conservation of Water Bodies

- Urban water bodies like lakes, tanks, and ponds also play a very important role in the management of urban flooding by reducing the stormwater run-off by capturing it.



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6. Role of Science and Technology

- The management of urban flooding has to be treated holistically in a multi-disciplinary manner.
- Science and technology can play a significant role in improved monitoring, modelling/ forecasting, and decision-support systems.
- One method for improving the preparedness for urban flooding is by setting up a vulnerability-based geospatial framework to generate and analyze different scenarios.
- It helps in identifying and planning for the most effective/ appropriate actions in a dynamic way to incorporate day-to-day changes that take place in urban areas, having the potential to alter the prevailing vulnerability profile.